

LCOE*: Re-thinking LCOE for Photovoltaic Systems

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Abstract — Solar industry is working towards reducing the levelized cost of energy (LCOE). The LCOE formulation includes various implicitly connected cost and physical design parameters. This entails complex cost analyses by economists, and electrical analysis and energy yield calculations of solar farms by device physicists/technologists. The standard formulation requires a constant exchange of information to optimize the design of a specific farm. In this paper, we present a fundamental reformulation of LCOE (in terms of LCOE*) to deconvolve the cost vs. energy yield analysis so that economists and technologists can work independently towards minimizing LCOE. We validate our LCOE* formulation by comparing LCOE calculated by the traditional vs. new method. We further provide a global analysis of LCOE* as an illustrative example. This LCOE* will accelerate new technology development (e.g. bifacial PV) by allowing a rapid and transparent analysis of its cost-performance trade-off.

Index Terms — levelized cost of energy (LCOE), photovoltaic systems, economic parameters, physical parameters.

I. INTRODUCTION

Levelized cost of energy (LCOE) is the widely accepted metric to compare and assess the economics of various sources of energy. For solar energy, LCOE refers to the ratio of the total cost of a PV system and the total energy yield over its lifetime [1]. Alternatively, it can be defined as the selling price per unit energy to break-even the cost price over the lifetime of a PV system. Estimating LCOE requires an integrated effort of PV economists and PV technologists. While economists deal with all the cost analyses, the technologists work on the physical design of a solar PV system for energy evaluation. In order to calculate and minimize LCOE, both economists and scientists require data from each other. For example, economists need the electrical and mechanical specifications – power capacity of the farm, module technology, inverter, racks, lifetime etc. – from the scientists, whereas scientists need available land area (/cost), subsidies and taxes, bank discount rate etc. – from economists to perform the same LCOE calculation. In simple terms, the LCOE formula is an implicit function of various cost and physical design parameters of a PV system. Due to this implicit nature of the formula, the economists and the technologists require a constant exchange of information to calculate LCOE for a farm or for a region, see Fig. 1(left). Hence, there is a need to decouple the costs and energy yield aspects of the LCOE formula to provide a simpler, more transparent, and physically-intuitive perspective of LCOE such that it is easier for economists and scientists to work in parallel towards the goal of minimizing LCOE.

In this paper, we achieve this objective of decoupling the economic and physical design parameters to facilitate

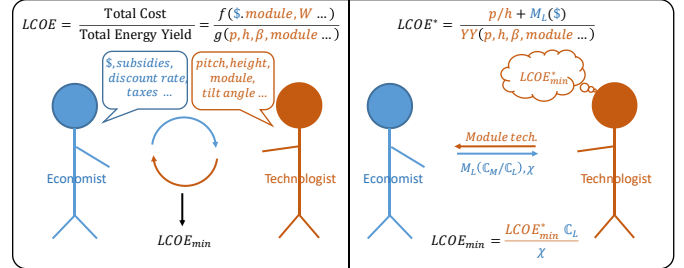


Fig. 1. The schematic shows how LCOE* formulation eases the exchange of data and optimization of design for minimizing LCOE.

simultaneous assessment of costs by economists and energy yield by technologists. In a fundamentally different perspective, we reformulate LCOE in terms of what we call LCOE*. We reparameterize the costs in order to separate the cost parameter, named cost ratio (M_L), and physical design parameters such as pitch, height, tilt angle, albedo etc. Fig. 1(right) schematically portrays how the LCOE* helps decouple the functions of an economist and a technologist and how it differs from traditional LCOE calculation involving an iterative communication (Fig. 1(left)). This way, after receiving the cost parameter (M_L) from the economist, the technologist can work independently on the design optimization of a solar PV system towards reducing the LCOE. Our results validate the LCOE* formulation by comparing the LCOE calculated by our approach to the traditional LCOE published in the literature.

In Sec. II, we present the complete parameterization and formulation of LCOE* and provide the practical values of some parameters. Then, in Sec. III, we display and discuss our results which are followed by the summary of this work and our concluding remarks in Sec. IV.

II. REFORMULATION AND REINTERPRETATION OF LCOE

As mentioned earlier, LCOE is defined as the ratio of the total cost of a PV system and the total energy yield of the system over its lifetime [1], i.e.,

$$\begin{aligned} LCOE &= \frac{\text{Total Cost (\$)}}{\text{Total Energy Yield (kWh)}} \\ &= \frac{C_{sys}(Y=0) + (\sum_{k=1}^Y C_{om}(k)) - C_{rv}(Y)}{E(Y)} \end{aligned} \quad (1)$$

where $C_{sys}(Y=0)$ is the fixed cost paid once during installation (i.e., $Y=0$) of the system. This includes the cost of modules ($c_{m,0}$), the cost of land ($c_{l,0}$), and the balance of system

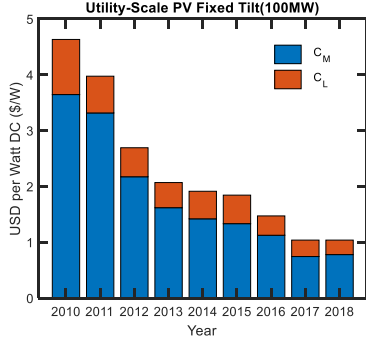


Fig. 2. Historical trend in LCOE [2] with costs that are regrouped into effective module cost (C_M) and effective land cost (C_L).

cost ($c_{bos,0}$) such as labor, permit, racks, inverters, etc. The recurring operations and maintenance cost (C_{om}) scales with the cost of maintaining individual modules ($c_{om,m}$) and the cost of maintaining the land ($c_{om,l}$). Finally, C_{rv} is the residual value of the modules ($c_{rv,m}$), the land ($c_{rv,l}$), and the equipment to be regained when the farm is decommissioned. C_{om} and C_{rv} are a function of the lifetime (number of years, Y) for which the solar farm is operated.

The total cost (numerator) in the LCOE expression in Eq. (1) can be *equivalently* written as the sum of effective module cost (C_M), effective land cost (C_L), and a fixed balance of system cost ($C_{bos,f}$), as shown in the following equation.

$$LCOE \equiv (C_M + C_L + C_{bos,f})/E(Y) \quad (2)$$

where

$$\begin{aligned} C_M &= c_{m,0} + \left(\sum_{k=1}^Y (c_{om,m}(k)(1+r)^{-k}) \right) \\ &\quad - C_{rv,m}(Y)(1+r)^{-Y} \\ C_L &= c_{l,0} + \left(\sum_{k=1}^Y (c_{om,l}(k)(1+r)^{-k}) \right) \\ &\quad - C_{rv,l}(Y)(1+r)^{-Y} \end{aligned} \quad (3)$$

Subscripts m , l , and bos stand for module, land, and balance of systems. Moreover, r is the discount rate that normalizes future costs in terms of current cost in order to have a fair comparison using a single metric.

The key insight of Eq. (2) is that the costs associated with a solar farm reflect two essential costs, namely, effective module cost (C_M) and effective land cost (C_L). The costs that scale with the number of modules (e.g., module size, material cost, racking, wiring in panels, inverters etc.) are included in C_M . Those costs that vary with the area of land (e.g., cost of land, fencing, cost for land curing/sculpting, etc.) are collected in C_L . Since the (typically small) balance of system cost (e.g. permit

cost) scales neither with the number of modules nor with the land, we collect them into the residual cost, $C_{bos,f}$.

The denominator of Eq. (2) describes the total energy yield $E(Y)$ of the solar farm:

$$E(Y) = \sum_{k=1}^Y E_0 (1-d)^k (1+r)^{-k} \quad (4)$$

Here, $E_0 = P_0 \times T_Y$ is the energy output for the first year expressed as the product of first year power output (P_0) and total number of active hours in a year (T_Y). The yearly energy degradation rate (d) defines the lifetime (Y , in years) of the solar farm. The discount rate (r) accounts for the following fact: If we continue to sell a unit of energy for c \$/watt, the present value of the lifetime revenue must account for the fact that future earnings are less valuable than present earning.

Given the dimensions of a solar farm and the modules installed, LCOE expression in Eq. (2) can be simplified even further, i.e.,

$$LCOE = \frac{C_M(r) \cdot h \cdot M \cdot Z + C_L(r) \cdot p \cdot M \cdot Z + C_{bos,f}}{YY(p, h, \beta, \gamma, R_A) \cdot M \cdot Z \cdot h \cdot \chi(d, r)} \quad (5)$$

where C_M is the cost per unit meter of the module (height), C_L is the cost per unit meter of land (pitch), M is the number of rows/arrays of modules and Z is the number of modules in an array. $YY(=E_0)$ is the yearly energy yield per meter of a pristine module for one period/pitch (p) such that the yearly energy of the farm is $E(Y) = YY \cdot M \cdot Z \cdot h \cdot \chi(d, r)$, where $\chi = \sum_{k=1}^Y (1-d)^k (1+r)^{-k}$. YY is a function of the physical design parameters (p, h, β, γ, R_A). Further, the residual cost associated with the balance of system ($c_{bos,f}$) is negligible compared the essential costs, C_M and C_L [2]. With these considerations, we arrive at the ‘essential leveled cost of energy’ ($LCOE^*$), as follows:

$$LCOE^* \equiv \frac{LCOE \cdot \chi}{C_L} = \frac{C_M/C_L + p/h}{YY(p, h, \beta, \gamma, R_A)} = \frac{p/h + M_L}{YY} \quad (6)$$

Eq. (6) defines an important design parameter $M_L (\equiv C_M/C_L)$ as the ratio of cost of module per unit length (height) over cost of land per unit length (pitch). Note that Eq. (6) expresses $LCOE$ explicitly in terms of the physical and cost parameters. The parametric reformulation of $LCOE$ in terms of $LCOE^*$ decouples the cost analysis (embedded in M_L) from design considerations (reflected in p/h and YY). In other words, we can optimize a farm with M_L as a parameter, realizing that M_L will evolve as the process technology evolves. Equation (6) shows that $LCOE$ is proportional to $LCOE^*$, since C_L and χ are location-specific constants. Therefore, minimizing $LCOE$ is equivalent to minimizing $LCOE^*$ for a given location.

Estimating C_M, C_L, χ . To find $LCOE$, we need to determine the practical values of C_M, C_L , and χ . In this regard, we refer to the cost benchmark report by NREL [2], which provides a detailed cost dissection for residential, commercial, and utility-scale PV systems. Fig. 2 shows the historical cost benchmark summary

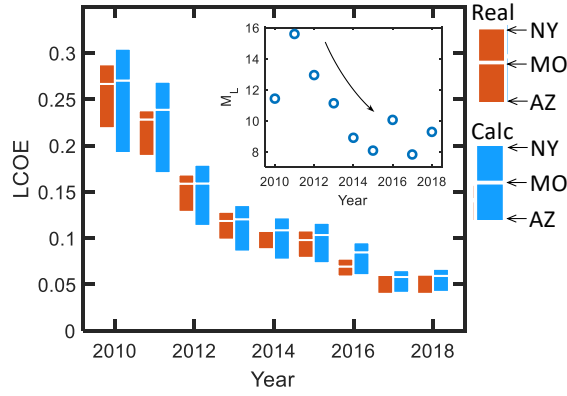


Fig. 3. Calculated LCOE vs. standard (real) LCOE for three locations. (Inset) Yearly trend shows that M_L ranges from 7 – 16 based on [2].

for fixed-tilt utility-scale PV. According to the definitions of C_M and C_L , we regroup these cost categories into the effective module cost (module cost, inverter, Hardware BOS, and soft costs – install labor) and effective land cost (soft costs – others). Since Fig. 2 costs are expressed in Dollars per watt (\$/W) while C_M and C_L involve Dollars per meter (\$/m), therefore

$$C_M = C_M W_h \frac{\$}{m}; C_L = C_L W_p \frac{\$}{m}; M_L = C_M W_h / C_L W_p \quad (7)$$

Here W_h and W_p are W/m of module height and W/m of land, respectively, of a solar farm. W_h is the product of cell efficiency (fraction) and the input power (1000 W/m^2), which is equal to 191 W/m assuming 1 m width of the module. For W_p , we follow the rule that 1 MW of plant requires 4 acres (4046.86 m^2) of land [3]. For 100 MW fixed-tilt plant (Fig. 2), $W_p = 100 \text{ MW} / (400 * 4046.86 \text{ m}) = 61.77 \text{ W/m}$, assuming unit width of the land. Taking degradation rate $d = 0.7\%$, bank discount rate $r = 6.5\%$ and farm lifetime $Y = 25 \text{ years}$, we calculate $\chi = 12.4$. The yearly energy yield is calculated using the farm models in [4]. We can now estimate $LCOE$ using the values of C_M , C_L , χ , and YY to compare it with real $LCOE$ values, as explained in the next section.

III. RESULTS

To validate our formulation of $LCOE^*$, let us compare the calculated $LCOE$ with the standard (or real) $LCOE$ for three locations – Phoenix, AZ, Kansas City, MO and New York City, NY [2]. Since the costs in Fig. 2 represent the average values across USA, thus the M_L , C_L and χ values are presumed to be the same for all the locations. However, the yearly energy yields differ substantially and is reflected in their respective $LCOE$. The overall agreement is excellent, the slight difference is attributed to the local variations in M_L , d , and Y .

With $LCOE^*$ formulation validated, let us analyze how $LCOE^*$ varies with M_L , locally and globally. The results are plotted in Fig. 4. In Fig. 4(a), we observe that a threshold value

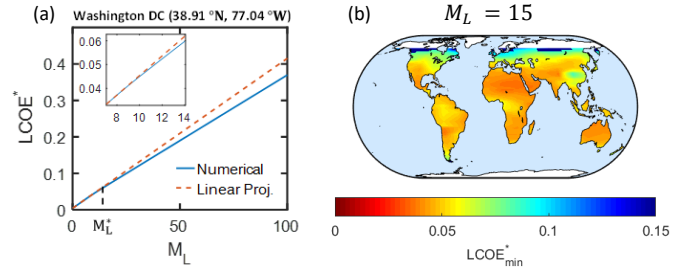


Fig. 4. (a) Non-linear variation of $LCOE^*$ with M_L . (b) Global $LCOE^*$ map for $M_L = 15$. Higher energy yield implies lower $LCOE^*$.

of M_L^* separates the linear and non-linear behavior of $LCOE^*$ with M_L . The separation demarcates the dominance of yearly energy yield (Eq. (6)) before M_L^* and that of cost (M_L) after M_L^* . The critical cost ratio M_L^* (typically 0-10) is estimated numerically for each location. The global map in Fig. 4(b) shows $LCOE^*$ values for $M_L = 15$. As expected, the locations with higher energy yield achieve lower $LCOE^*$ values. Thus, an estimation of M_L for each location around the world combined with respective χ values can be used to calculate the $LCOE$ global map.

IV. CONCLUSIONS

To summarize, we have reparametrized the complex LCOE formula into two independent/explicit components of cost and energy. We then reformulated and reinterpreted $LCOE$ in terms of effective $LCOE^*$. Using practical values of costs, degradation rate, discount rate, lifetime, and yearly energy yield, we calculated $LCOE$ based on the $LCOE^*$ formulation. Our results reproduce the historical trends of $LCOE$ calculated by the traditional methods. Thus validated, we further present the variation of $LCOE^*$ with cost ratio (M_L), locally and globally. Once we know the local effective land costs, we can easily predict $LCOE$ at any location in the world. This new approach to calculate $LCOE$ provides a transparent and powerful technique to assess $LCOE$ of new technologies (e.g. perovskite) by decoupling economics and physics of a solar PV system.

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